Cut rank under substitution in natural deduction terms

The Open Logic Project contributors

openlogic:proof-theory/propositions-as-types:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:subst (620-637); the proof span 638-663 is the author proof environment that immediately follows it.

Editorial note. The source statement prints a stray closing parenthesis after len(A); the unit is reproduced without mathematical changes, so the stray parenthesis is kept as authored.

Editorial scope. One original result of the Open Logic Project unit proof-theory/propositions-as-types, counted once: the cut rank of a substituted deduction. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Cut rank of a deduction. The cut rank of a natural deduction term measures the size of its largest redex, computed from the lengths of the formulas at the elimination-introduction pairs; the cut rank of a variable is the length of its type.. The lemma compares the cut rank of the substituted deduction with that of M and with the length of A (620-637).
- Substitution of deductions. Substituting a deduction N of A for the assumption variable x in M replaces every free occurrence of x in M by N.. The proof inspects how redexes can be created by this substitution, case by case (638-663).
- Redex in natural deduction terms. A redex is an elimination immediately applied to an introduction of the matching connective, such as a conjunction elimination acting on a conjunction introduction.. The four enumerated cases classify exactly which introduction inside the substituted term meets which elimination (622-631).

Lemma 0.1. *If $M[:N!A/:x!A]$ is a redex and $M \not\equiv x$, then one of the following cases holds:*

1. *M is itself a redex, or*
2. *M is of the form $\mathrm{and}_E ix$, and N is of the form $\mathrm{and}_I P_1 P_2$*

3. *M is of the form* $\mathrm{or}_E i x_1 P_1 x_2 P_2$*, and N is of the form* $\mathrm{or}_I i Q$

4. *M is of the form xQ, and N is of the form $\lambda[x][P]$*

In the first case, $\mathrm{cr}\, M[N/x] = \mathrm{cr}\, M$*; in the other cases,* $\mathrm{cr}\, M[N/x] = \mathrm{len}\, !A$*).*

Proof. Proof by induction on M.

1. If M is a single variable y and $y \not\equiv x$, then $y[N/x]$ is y, hence not a redex.

2. If M is of the form $\mathrm{and}_I N_1 N_2$, or $\lambda[x][N]$, or $\mathrm{or}_I i !AN$, then $M[:N!A/:x!A]$ is also of that form, and so is not a redex.

3. If M is of the form $\mathrm{and}_E i P$, we consider two cases.

   (a) If P is of the form $\mathrm{and}_I P_1 P_2$, then $M\,\mathrm{id}\,\mathrm{and}_E i \mathrm{and}_I P_1 P_2$ is a redex, and clearly

   $$M[N/x]\,\mathrm{id}\,\mathrm{and}_E i \mathrm{and}_I P_1[N/x] P_2[N/x]$$

   is also a redex. The cut ranks are equal.

   (b) If P is a single variable, it must be x to make the substitution a redex, and N must be of the form $\mathrm{and}_I P_1 P_2$. Now consider

   $$M[N/x]\,\mathrm{id}\,\mathrm{and}_E i x[\mathrm{and}_I P_1 P_2/x],$$

   which is $\mathrm{and}_E i \mathrm{and}_I P_1 P_2$. Its cut rank is equal to $\mathrm{cr}\, x$, which is $\mathrm{len}\, !A$.

The cases of $\mathrm{or}_E N x_1 N_1 x_2 N_2$ and PQ are similar. □

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